

HelixCode Feature Status

HelixCode — Feature Status

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Comprehensive per-feature status ledger (§11.4.153) for the **HelixCode** stack at release `codeserver-1.0.0-dev-0.0.3`. Every user-facing capability is enumerated with the §11.4.153 field set — Component / Feature / Category / Implementation (the real file) / Wiring (how it is genuinely end-user-reachable, §11.4.108) / Real-use / Tests-coverage (§11.4.169) / Validation / Video-confirmation. Validation vocabulary (§11.4.45): **PASS · SKIP · OPERATOR-BLOCKED · OPERATOR-ATTENDED · PENDING**. Autonomous captured evidence lives under the git-ignored `qa-results/` tree (regenerated by re-running the suite, §11.4.77); the release-gate aggregate is `qa-results/run_all/20260701T151300Z-3182121/` (`summary.txt`), with per-suite evidence under `qa-results/tests/<suite>/`.

Reconciliation basis (§11.4.6). This ledger reconciles against `README.md`, `docs/guides/AUTH.md`, and `docs/changelogs/codeserver-1.0.0-dev-0.0.3.md`. The real-account SSH-key auth surface is **LIVE-validated** this release — release-gate run = **23/23 suites PASS, 0 FAIL** (`summary.txt`), Go gate = **70 tests, -race, 81.8% coverage**. Rows that were "in progress / PENDING" in Revision 1 are now PASS with cited captured evidence.

Components

- **Caddy edge** — the only containerized component (rootless Podman, §11.4.76/§11.4.161). Terminates TLS on :52443 (HTTP :52080→HTTPS 301), speaks HTTP/3 (QUIC), compresses with Brotli (custom xcaddy build), gates every request via `forward_auth` → `helix-auth`, blocks `/proxy`, strips client X-Helix-User. `deploy/Caddyfile`, `deploy/caddy/Dockerfile`.
- **helix-auth** — host-native **Gin** forward-auth gate (`systemd --user, loopback 127.0.0.1:8081, non-root`). Issues + verifies the SSH-key challenge (`ssh-keygen -Y verify` vs the account's

authorized_keys), issues the signed session cookie, **fails closed**, stores no password. services/auth_gate/.

- **code-server** — browser VS Code, **host-native as the real user** (milosvasic), --auth none on loopback 127.0.0.1:8080 (auth enforced upstream at the gate/edge), projects served read-write, Explorer defaults to \$PROJECTS_ROOT. deploy/systemd/helix-code-server.service.
- **Operator scripts** — install / setup / start / stop / status / doctor + install-auth.sh (tarball code-server + units), install-edge-boot.sh (edge Quadlet), harden-loopback.sh (UID-scoped loopback firewall).
- **Boot-survival** — rootless systemd --user units (linger) + the edge Quadlet.

Feature ledger (§11.4.153)

Component	Feature	Category	Implementation
Caddy edge	TLS HTTPS edge (self-signed LAN, per-boot)	edge/tls	deploy/Caddyfile, deploy/c Dockerfile
Caddy edge	HTTP→HTTPS redirect (:52080→:52443, 301)	edge/tls	deploy/Caddyfile
Caddy edge	HTTP/3 (QUIC) + Brotli compression	edge/perf	deploy/Caddyfile (encode zstd br gzip), deploy/caddy/Dockerfile (xcaddy-)
Caddy edge	forward_auth gate wiring (every req → helix-auth /auth)	edge/auth	deploy/Caddyfile forward_a
Caddy edge	/proxy blocked (403) + client X-Helix-User stripped	edge/security	deploy/Caddyfile
Caddy edge	Let's Encrypt ACME issue/renew/rotation (local Pebble CA)	edge/tls	deploy/.env TLS_MODE=letsencrypt, deploy/Caddyfile
Caddy edge	Real public Let's Encrypt (public domain / DNS-01)	edge/tls	docs/guides/TLS.md
Caddy edge	Edge reboot-persistence (rootless Quadlet)	edge/boot	deploy/quadlet/helixcode-caddy.container, scripts/install-edge-boot.sh
helix-auth		auth	

Component	Feature	Category	Implementation
	SSH-key challenge-response login (verify vs <code>authorized_keys</code> , no password, nothing stored)		<code>services/auth_gate/</code> (<code>challenge.go</code> , <code>verifier.go</code> , <code>cookie.go</code> , <code>server.go</code>), <code>deploy/systemd/helix-auth.service</code>
helix-auth	Login redirect for browser navigations (<code>303→/login</code> ; non-HTML/non-GET → 401)	auth	<code>services/auth_gate/server</code> <code>handleAuth</code> (commit 75e2d9)
helix-auth	Auth fails CLOSED (gate down/errored → denied, never bypassed)	auth/security	<code>services/auth_gate/server</code> <code>deploy/Caddyfile</code>
helix-auth	Hardened session cookie (<code>__Host-</code> + HMAC + <code>HttpOnly</code> + <code>Secure</code> + <code>SameSite=Strict</code> , regen on login)	auth/security	<code>services/auth_gate/cookie</code>
helix-auth	Rate-limit + CSRF double-submit + verify-spawn ceiling (DoS-hardened)	auth/security	<code>services/auth_gate/rateli</code> <code>server.go</code>
helix-auth	Security hardening (server-pinned principal CVE-2026-35414, key-type allow-list, OpenSSH floor, exec/temp hygiene)	auth/security	<code>services/auth_gate/verifi</code> <code>sshversion.go</code> , <code>config.go</code>
helix-auth	Login-form copy/paste clipboard buttons (accessible, graceful in insecure context)	auth/ux	<code>services/auth_gate/assets</code> <code>login_enhance.js</code>
code-server	Host-native editor served (authed 200, x-Helix-User: milosvasic)	editor	<code>deploy/systemd/helix-code</code> <code>server.service</code>
code-server	Tarball install (<code>~/.local/lib</code> , non-root; npm fallback), pinned 4.117.0	editor/install	<code>scripts/install-auth.sh</code>
code-server	Projects served read-write; Explorer defaults	editor	

Component	Feature	Category	Implementation
	to \$PROJECTS_ROOT (convenience view, NOT a jail)		deploy/systemd/helix-code server.service, deploy/.en PROJECTS_ROOT
code-server	SSH-key git from the integrated terminal (host ~/.ssh)	editor/env	host-native as real user
code-server	.bashrc / profile sourced in fresh terminal	editor/env	host-native as real user
code-server	Extension (plugin) install + use from Open VSX marketplace	editor/ extensions	tests/types/extensions_au bank tests/banks/helixcod extensions.yaml
code-server	In-browser editing (type/save via UI, file tree, command palette)	editor/ui	code-server workbench
code-server	UI-driven marketplace test (browse → install → use through the editor UI)	editor/ui	(planned driver)
code-server	file-watcher fix (files.watcherExclude seeded to volume)	editor/env	persistent settings volum
Operator scripts	Loopback firewall (UID-scoped DROP to :8080 for non-account UIDs)	security/host	scripts/harden-loopback.s
Operator scripts	Boot-survival systemd -- user units (linger, WantedBy=default.target)	boot	scripts/install-auth.sh, d systemd/*.service
Stack	Resilience (flood / kill- recover / fd pressure)	resilience	tests/types/ {load_auth,stress_chaos_a
Stack	Performance / memory bounds (p50/p95/p99, RSS < 60%)	performance/ memory	tests/types/ {benchmark_auth,memory_au

Component	Feature	Category	Implementation
Stack	No secret leak / rootless	security	tests/types/security_auth

Release-gate run (§11.4.169)

Aggregate qa-results/run_all/20260701T151300Z-3182121/summary.txt
— **23/23 PASS, 0 FAIL:**

- **Auth surface (LIVE, real id_ed25519):** e2e_auth 5/5 · security_auth 5/5 · challenges_auth 5/5 · concurrency_auth 5/5 · helixqa_auth 4/0/1 · load_auth 3/0/1 · memory_auth 2/2 · benchmark_auth 4/4 · stress_chaos_auth 5/0/1 (live kill+recover, fail-closed verified).
- **Core:** unit 5/5 · race 2/2 (-race) · load 3/3.
- **Go gate: 70 tests, -race, 81.8% coverage.**
- **Legacy (retired container+password model):** integration, e2e, security, tls_letsencrypt, full_automation, concurrency, memory, benchmark, helixqa, challenges each report an honest **SKIP** (topology_unsupported, "superseded by <name>_auth") — no false-FAIL (§11.4.1).
- **Extension surface (separate live run):** extensions_auth 5/5 checks captured (qa-results/tests/extensions_auth/20260701T162817Z-4080597).

Honest boundaries (§11.4.6)

- **No §11.4.153 real-use VIDEO exists yet** for any HelixCode feature — every row's **Video-confirmation is PENDING**. The autonomous captured evidence backing each PASS is the qa-results/** log + .txt artefact cited in the row (§11.4.69/§11.4.107), NOT a recorded video. Producing the per-feature real-use videos is a tracked gap; nothing here is claimed video-confirmed until the recording lands.
- **Editor "jail" is cosmetic.** The Explorer defaults to \$PROJECTS_ROOT, but the integrated terminal, **File > Open Folder**, and extensions retain **full real-user (milosvasic) host access by design** — a convenience view, not a security boundary. Real isolation would need a container / VM / chroot (out of scope per operator decision). Defense-in-depth (deploy stream): --disable-workspace-trust
 - /proxy/ blocked at the edge.
- **--auth none on loopback:8080 residual.** code-server runs unauthed on loopback; all auth is at the gate/edge. A local process reaching 127.0.0.1:8080 bypasses the gate — mitigated

by scripts/harden-loopback.sh (a UID-scoped firewall rule the operator applies as root; the project never uses sudo).

- **Full in-browser editor interaction** (typing, file tree, command palette) is **OPERATOR-ATTENDED** until a browser-automation adapter lands (HCA-QA-UI-001); every HTTP/CLI-drivable journey is autonomously verified.
- **In-flight (subagents, NOT done):** the login-form clipboard copy/paste buttons (login_enhance.js — source landed, unvalidated) and the UI-driven marketplace test are **PENDING** — the current extension coverage is the HTTP/CLI-driven extensions_auth suite (live 5/5).
- **Real, publicly-trusted Let's Encrypt** cannot be autonomously proven on a LAN box; the ACME code path is present (TLS_MODE=letsencrypt), the legacy local-CA suite now SKIPS as superseded, and the operator enables the public CA per docs/guides/TLS.md.
- **DOCX export.** Per §11.4.153(6) this doc class ships HTML + PDF + **DOCX**. This release regenerates all three for Status.md + Status_Summary.md via pandoc; the project has **no standing DOCX pipeline** in tests/run_all_types.sh yet — wiring a §11.4.60/§11.4.65 auto-export step for DOCX is a tracked follow-up.